INCH-POUND

MIL-M-14/5
AMENDMENT 1
7 November 1990

MILITARY SPECIFICATION SHEET

MOLDING COMPOUNDS, DIALLYL ISO-PHTHALATE THERMOSETTING, CONTAINING MINERAL FILLERS, GLASS OR POLYMERIC FILLERS

This amendment forms a part of MIL-M-14/5, dated 2 April 1990, and is approved for use by all Department and Agencies of the Department of the Defense.

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Table I: Delete and substitute:

TABLE I. *Qualification test requirements for diallyl iso-phthalate resin molding compounds, mineral/glass filled.*

Requirement	Type GII-30	Type GII-30F	Type MIG	Type MIG-F	Type SIG	Type SIG-F
			Mechanical/physical			
Compressive strength, endwise	24,000	24,000	20,000	20,000	24,000	24,000
Dimensional stability	0.1	0.1	0.2	0.2	0.2	0.2
Flexural strength	13,000	13,000	10,000	10,000	13,000	13,000
Heat deflection temperature	230	230	200	200	230	230
Heat resistance	170	170	140	140	160	160
Impact strength, side[1]	3.0	3.0	0.35	0.35	0.50	0.50
Tensile strength	6,000	6,000	6,000	6,000	6,000	6,000
Water absorption	0.5	0.5	0.7	0.7	0.5	0.5

See footnote at end of table.

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Table I: Delete and substitute (page 3 attached):

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Table II: Delete and substitute (page 4 attached):

Custodians:
Army - ME
Navy - SH
Air Force - 20

Review activities:
Army - ER, MI, MR, AR, EA
Navy - AS, OS
DLA - GS

User activities:
Navy - YD, EC

Preparing activity:
Navy - SH
(Project 9330-1170)

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TABLE I. *Qualification test requirements for diallyl iso-phthalate resin molding compounds, mineral/glass filled – Continued.*

Requirement	Type GII-30	Type GII-30F	Type MIG	Type MIG-F	Type SIG	Type SIG-F
			Electrical			
Arc resistance	115	115	115	115	115	115
Dielectric breakdown:						
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)						
Step-by-step test	45	45	45	45	45	45
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)						
Step-by-step test	40	40	45	45	40	40
Dielectric constant:						
at 1 kilohertz	4.6	4.6	5.0	5.0	4.6	4.6
	4.7	4.7	5.2	5.2	4.7	4.7
at 1 megahertz	4.6	4.6	5.0	5.0	4.6	4.6
	4.7	4.7	5.2	5.2	4.7	4.7
Dielectric strength:						
Short-time test	325	325	325	325	325	325
Step-by-step test	300	300	300	300	300	300
Short-time test	325	325	325	325	325	325
Step-by-step test	300	300	300	300	300	300
Dissipation factor:						
at 1 kilohertz	0.015	0.015	0.019	0.019	0.010	0.010
	0.018	0.018	0.020	0.020	0.013	0.013
at 1 megahertz	0.020	0.020	0.025	0.025	0.020	0.020
	0.023	0.023	0.025	0.025	0.023	0.023
Surface resistance	5,000	5,000	5.0	5.0	10	5,000
Volume resistance	5,000	5,000	2.0	2.0	10	5,000
Water extract conductors	60×10^{-6}	60×10^{-6}	–	–	–	–
			Combustion			
Flame resistance:						
Ignition time	–	90	–	90	–	90
Burning time	–	90	–	90	–	90
Flammability/Thickness-Inch	–	V-0 / 1/8	–	V-0 / 1/16	–	V-0 / 1/16
Toxicity when heated:						
Carbon dioxide		15,000		15,000		
Carbon monoxide		1,000		1,000		
Ammonia		2,500		2,500		
Aldehydes as HCHO		50		50		
Cyanide as HCN		60		60		
Oxide of nitrogen at NO_2		100		100		
Hydrogen chloride		100		100		

[1]The side of the test specimen is that area formed by the chase of the mold.

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TABLE II. *Batch acceptance test requirements for diallyl iso-phthalate molding compounds, mineral/glass filled.*

Property to be tested	Type GII-30	Type GII-30F	Type MIG	Type MIG-F	Type SIG	Type SIG-F
	Performance requirement for each compound.					
Arc resistance	115	115	115	115	115	115
Dielectric constant at 1 megahertz	4.7	4.7	5.2	5.2	4.7	4.7
Dielectric strength, step-by-step	300	300	300	300	300	300
Dissipation factor at 1 megahertz	0.023	0.023	0.025	0.025	.023	0.023
Flexural strength	13,000	13,000	10,000	10,000	13,000	13,000
Impact strength, side[1]	3.0	3.0	0.35	0.35	0.50	0.50
Water absorption	0.5	0.5	0.7	0.7	0.5	0.5
Water extract conductors	60×10^{-6}	60×10^{-6}	–	–	–	–

[1]The side of the test specimen is that area formed by the chase of the mold.

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TABLE II. *Batch acceptance test requirements for diallyl ortho-phthalate molding compounds, mineral, glass or polymer fiber filled.*

Property to be tested	Type GDI-30	Type GDI-30F	Type GDI-300	Type DGI-300F	Type MDG	Type MDG-F	Type SDG	Type SDG-F	Type SDI-5	Type SDI-30
	Performance requirement for each compound.									
Arc resistance	115	115	115	115	115	115	115	115	100	115
Dielectric constant at 1 megahertz	4.7	4.7	5.0	5.0	5.0	5.0	4.7	4.7	3.9	3.9
Dielectric strength, step-by-step	300	300	300	300	300	300	300	300	300	300
Dissipation factor at 1 megahertz	0.023	0.023	0.030	0.030	0.030	0.030	0.023	0.023	0.025	0.025
Flexural strength	13,000	13,000	35,000	35,000	10,000	10,000	13,000	13,000	8,000	10,000
Impact strength, side[1]	3.0	3.0	30.0	30.0	0.30	0.30	0.50	0.50	0.60	3.0
Water absorption	0.5	0.5	0.5	0.5	0.7	0.7	0.5	0.5	0.5	0.5
Water extract conductance	50×10^{-6}	50×10^{-6}	–	–	–	–	–	–	–	–

[1]The side of the test specimen is that area formed by the chase of the mold.

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Air Force – 20

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DLA – GS

User activities:
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Preparing activity:
Navy - SH
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